

Lecture 13 Worksheet

MemRef, Layout, Memory Space

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Exercise A. MemRef type anatomy annotation

```
memref<64x128xf16, strided<[160, 1], offset: ?>, #npu.mem<"sram", bank=1, align=64>>
```

Field	Answer
Shape/rank	
Element type	
Layout / strides	
Dynamic metadata	
Memory space contract	
NPU verifier rule	

Exercise B. Subview offset/stride 계산

Base: memref<1024x1024xf16, strided<[1024,1], offset:0>, 0>. Tile: %A[%m0,%k0][64,128][1,1].

Item	Answer
result offset	
result sizes	
result strides	
copy or view?	
DMA descriptor fields	

Exercise C. MatMul memory-space mapping

Buffer	Shape	DType	Space	Bank/Align	Lifetime	Reason
A_tile ping	64x128	f16				
A_tile pong	64x128	f16				
B_tile ping	128x64	f16				
B_tile pong	128x64	f16				
C_acc	64x64	i32				
C_stage	64x64	i8				

Exercise D. Footprint 계산

Component	Formula	Bytes
A_tile	64*128*2	
B_tile	128*64*2	
Ping-pong A/B	2*(A+B)	
C_acc	64*64*4	
C_stage	64*64*1	
SRAM total excluding C_acc	ping-pong A/B + C_stage + padding/tags	

Exercise E. Runtime descriptor fields

Field	Required?	Reason
base/handle		
byte offset		
rank		
sizes[]		
byte strides[]		
dtype		
memory_space		
layout_id		
alignment/flags		

Exercise F. FileCheck test

```
// CHECK-LABEL: func.func @kernel
// TODO: check SRAM allocation
// CHECK-NOT: memref.copy
```